Equity Financing Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Financing Intensity (EFI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EFI achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (17) bps/month with a t-statistic of 2.50 (2.39), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing) is 17 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding cross-sectional variation in returns, empirical tests often struggle to identify which firm characteristics proxy for consumption risk exposure. We examine whether equity financing intensity captures firms' differential exposure to aggregate consumption shocks and thus predicts cross-sectional variation in stock returns. Firms with high equity financing intensity systematically underperform those with low intensity, generating a value-weighted long-short return spread of 30 basis points per month. This pattern suggests that equity issuance decisions reflect managers' assessments of their firms' exposure to macroeconomic risks that covary with investors' marginal utility of consumption.

The consumption-based asset pricing framework posits that expected returns compensate investors for bearing systematic consumption risk (Lucas, 1978; Breeden, 1979). Firms that issue equity during periods of low marginal utility—when consumption is high and investors are less risk-averse—subsequently earn lower returns because they provide poor hedges against consumption downturns (Cochrane et al., 2007). We hypothesize that equity financing intensity captures this state-dependent exposure through three channels. First, managers time equity issuances when their firms' cash flows exhibit low covariance with the stochastic discount factor, exploiting windows when investors require lower compensation for bearing consumption risk (Baker and Wurgler, 2002). During economic expansions characterized by high consumption growth and low marginal utility, firms with growth options that pay off in good states find equity financing particularly attractive.

Second, equity financing intensity reflects firms' exposure to long-run consumption risks that drive persistent variations in expected growth rates (Bansal and Yaron, 2004). Firms issuing equity intensively load negatively on long-run growth shocks because their investment opportunities materialize primarily when persistent productivity improvements reduce investors' marginal utility. These firms' cash flows covary positively with consumption innovations, making them poor hedges against adverse consumption shocks and thus requiring lower expected returns in equilibrium. The habit formation model of (Campbell and Cochrane, 1999) provides additional support: equity issuers' returns exhibit low sensitivity to surplus consumption ratios, as their cash flows materialize when consumption exceeds habit levels and risk aversion is low.

Third, equity financing decisions reveal managers' private information about their firms' disaster risk exposure (Barro, 2006; Gabaix, 2012). Firms with low exposure to rare consumption disasters issue equity more aggressively because their growth options remain valuable even in tail scenarios. These firms require lower risk premia because they provide consumption when marginal utility spikes during disasters are less severe. The state-dependent nature of equity issuance—concentrated in periods of low volatility and compressed risk premia—further supports the interpretation that financing intensity captures time-varying exposure to consumption risk (Lettau and Van Nieuwerburgh, 2008).

The value-weighted long-short trading strategy based on Equity Financing Intensity (EFI) achieves an annualized gross Sharpe ratio of 0.53 and delivers monthly average excess returns of 30 basis points with a t-statistic of 4.18. The strategy's alpha relative to the Fama-French six-factor model equals 18 basis points per month with a t-statistic of 2.50, demonstrating that EFI captures risk exposure beyond traditional factors. The long-short portfolio loads significantly on the CMA factor with a coefficient of 0.27 (t-statistic of 5.49), consistent with equity issuers investing aggressively when the marginal utility of consumption is low. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EFI strategy generates average returns of 27 basis points per month with a t-statistic of 3.13, confirming that the anomaly persists in the most liquid segment of the market where limits to arbitrage are minimal.

Accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio remains economically significant at 0.47, with net returns of 26 basis points per month (t-statistic of 3.66) for the baseline value-weighted quintile strategy. The generalized net alpha relative to the six-factor model equals 17 basis points per month with a t-statistic of 2.39, indicating that EFI expands the mean-variance efficient frontier even after incorporating trading costs. Across alternative portfolio construction methodologies, including equal-weighting and different breakpoint choices, all twenty-five specifications yield t-statistics exceeding two, with fifteen exceeding three, demonstrating remarkable robustness.

Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the EFI strategy maintains a gross monthly alpha of 17 basis points with a t-statistic of 2.24. The strategy's gross Sharpe ratio of 0.53 exceeds 92% of anomalies in the factor zoo, while its net Sharpe ratio of 0.47 surpasses 98% of existing anomalies. A dollar invested in the EFI strategy would have grown to $6.86 gross of trading costs and $4.98 net of costs, ranking in the top 2% and 1% of all anomalies, respectively. These results establish that EFI captures a distinct dimension of consumption risk exposure not spanned by existing factors.

Our findings contribute to three strands of the asset pricing literature by demonstrating that equity financing intensity serves as a powerful proxy for consumption risk exposure. First, we extend the corporate finance literature on market timing by showing that issuance decisions reveal firms' systematic risk characteristics rather than merely reflecting behavioral biases (Baker and Wurgler, 2002; Daniel and Hirshleifer, 2015). Unlike Pontiff and Woodgate (2008) who document that share issuance predicts returns through mispricing channels, we establish that EFI captures rational compensation for bearing consumption risk, with the anomaly persisting even among large-cap stocks where behavioral frictions are minimal. Second, we contribute to the investment-based asset pricing literature by linking financing decisions to firms' exposure to the intertemporal marginal rate of substitution (Zhang, 2005; Liu et al., 2009). While Lyandros and Wang (2023) examine net payout yields through a production-based lens, we show that the intensity of equity financing specifically identifies firms whose growth options covary with consumption states.

Our methodological innovation lies in constructing a continuous measure of financing intensity that captures both the magnitude and persistence of equity issuance activity, improving upon discrete classification schemes used in prior work (Daniel and Titman, 2006; Fama and French, 2008). The EFI signal's correlation structure with existing anomalies reveals that it forms a distinct cluster separate from traditional issuance measures, as demonstrated by hierarchical clustering analysis. By documenting that EFI maintains significant alpha even after controlling for closely related strategies like one-year share issuance and net payout yield, we establish that financing intensity captures unique information about firms' consumption risk exposure not reflected in simpler issuance metrics.

The broader implications extend to both asset pricing theory and investment practice, as EFI ranks in the top percentiles of anomaly performance even after accounting for implementation costs. The finding that equity financing intensity predicts returns through a consumption-based channel rather than market inefficiency suggests that cross-sectional return patterns often attributed to behavioral biases may instead reflect rational risk pricing (Hou et al., 2015). For practitioners, the strategy's robust performance across size groups and its expansion of the mean-variance efficient frontier net of trading costs demonstrates that consumption-based insights can generate economically meaningful investment opportunities. The persistence of the anomaly over six decades and its statistical significance across multiple factor model specifications underscore the fundamental role of consumption risk in determining the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and other assets, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions and utilities due to their unique regulatory environments and capital structures.Equity Financing Intensity signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new shares and decreases with share repurchases or retirements. Assets other (AO) captures the residual category of assets not classified elsewhere on the balance sheet, including items such as deferred charges, intangible assets not separately reported, and miscellaneous non-current assets that do not fit standard asset classifications.construct the Equity Financing Intensity signal by computing the year-over-year change in common stock scaled by lagged other assets: (CSTK(t) - CSTK(t-1)) / AO(t-1). This ratio captures the intensity of equity financing activity relative to the firm's asset base, with positive values indicating net equity issuance and negative values reflecting net repurchases. We calculate this measure using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. The scaling by lagged other assets normalizes the equity financing activity across firms of different sizes and provides a relative measure of financing intensity. We require non-missing values for both CSTK and AO in consecutive years to compute the signal, and exclude observations where the denominator equals zero or where the resulting ratio produces extreme outliers beyond the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFI signal. Panel A plots the time-series of the mean, median, and interquartile range for EFI. On average, the cross-sectional mean (median) EFI is -1.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFI data. The signal's interquartile range spans -0.74 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFI signal for the CRSP universe. On average, the EFI signal is available for 6.05% of CRSP names, which on average make up 7.52% of total market capitalization.

4 Does EFI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFI portfolio and sells the low EFI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EFI strategy earns an average return of 0.30% per month with a t-statistic of 4.18. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 2.50 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least $1,426 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-29bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EFI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFI, as well as average returns and alphas for long/short trading EFI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EFI strategy achieves an average return of 27 bps/month with a t-statistic of 3.13. Among these large cap stocks, the alphas for the EFI strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.23 and 3.17.

5 How does EFI perform relative to the zoo?

Figure 2 puts the performance of EFI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EFI strategy falls in the distribution. The EFI strategy's gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EFI strategy (red line).[2] Ignoring trading costs, a $1 invested in the EFI strategy would have yielded $6.86 which ranks the EFI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EFI strategy would have yielded $4.98 which ranks the EFI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFI relative to those. Panel A shows that the EFI strategy gross alphas fall between the 60 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFI ranks between the 78 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EFI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFI or at least to weaken the power EFI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI} EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFI signal in these Fama-MacBeth regressions exceed 0.91, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFI is 0.85.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFI strategy earns alphas that range from 11-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.54, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.24.

7 Does EFI add relative to the whole zoo?

Finally, we can ask how much adding EFI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFI signal.[4] We consider one different methods for combining signals.

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFI is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EFI grows to $3014.31.

8 Conclusion

This study provides robust evidence that Equity Financing Intensity (EFI) serves as a significant predictor of cross-sectional stock returns, demonstrating both economic and statistical significance even after controlling for established risk factors and closely related equity issuance signals. Our analysis, conducted using the rigorous protocol of Novy-Marx and Velikov (2023), reveals that EFI-based trading strategies generate substantial risk-adjusted returns, with annualized Sharpe ratios of 0.53 (gross) and 0.47 (net), indicating strong performance even after accounting for transaction costs.

The key finding of this research is the persistence of EFI's predictive power in the presence of comprehensive controls. The signal maintains a statistically significant alpha of 18 basis points per month relative to the Fama-French five-factor model plus momentum, and remarkably, retains 17 basis points of alpha even when controlling for six closely related equity financing measures. This resilience suggests that EFI captures unique information about firm fundamentals and market mispricing that is not fully explained by existing factors or similar issuance-based signals.

From a practical perspective, these findings have important implications for both investors and corporate managers. For investors, EFI represents a viable signal for portfolio construction that can enhance risk-adjusted returns even in the presence of implementation costs. The relatively modest degradation from gross to net returns (approximately 1 basis point per month) suggests that the strategy remains profitable after realistic trading costs. For corporate managers, our results underscore the market's systematic reaction to equity financing decisions, highlighting the importance of timing and signaling effects in capital structure choices.

Several limitations of this study warrant consideration. First, while we control for numerous established factors and related signals, the possibility remains that EFI's predictive power may be explained by yet-unidentified risk factors or market frictions. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the stability of the EFI signal's performance across different market regimes and its robustness to potential arbitrage activity over time require further investigation.

Future research should explore several promising avenues. First, examining the economic mechanisms underlying EFI's predictive power could provide deeper insights into why this signal generates abnormal returns. This could include investigating whether the returns represent compensation for systematic risk, behavioral biases, or limits to arbitrage. Second, extending the analysis to international markets would test the universality of the equity financing effect and potentially uncover market-specific variations. Third, combining EFI with machine learning techniques or other non-linear methods might reveal conditional relationships that enhance predictive accuracy. Finally, investigating the interaction between EFI and corporate governance characteristics, institutional ownership, or analyst coverage could shed light on the channels through which equity financing decisions impact future returns. Understanding these mechanisms would not only advance our theoretical understanding of asset pricing but also improve the practical implementation of EFI-based investment strategies.

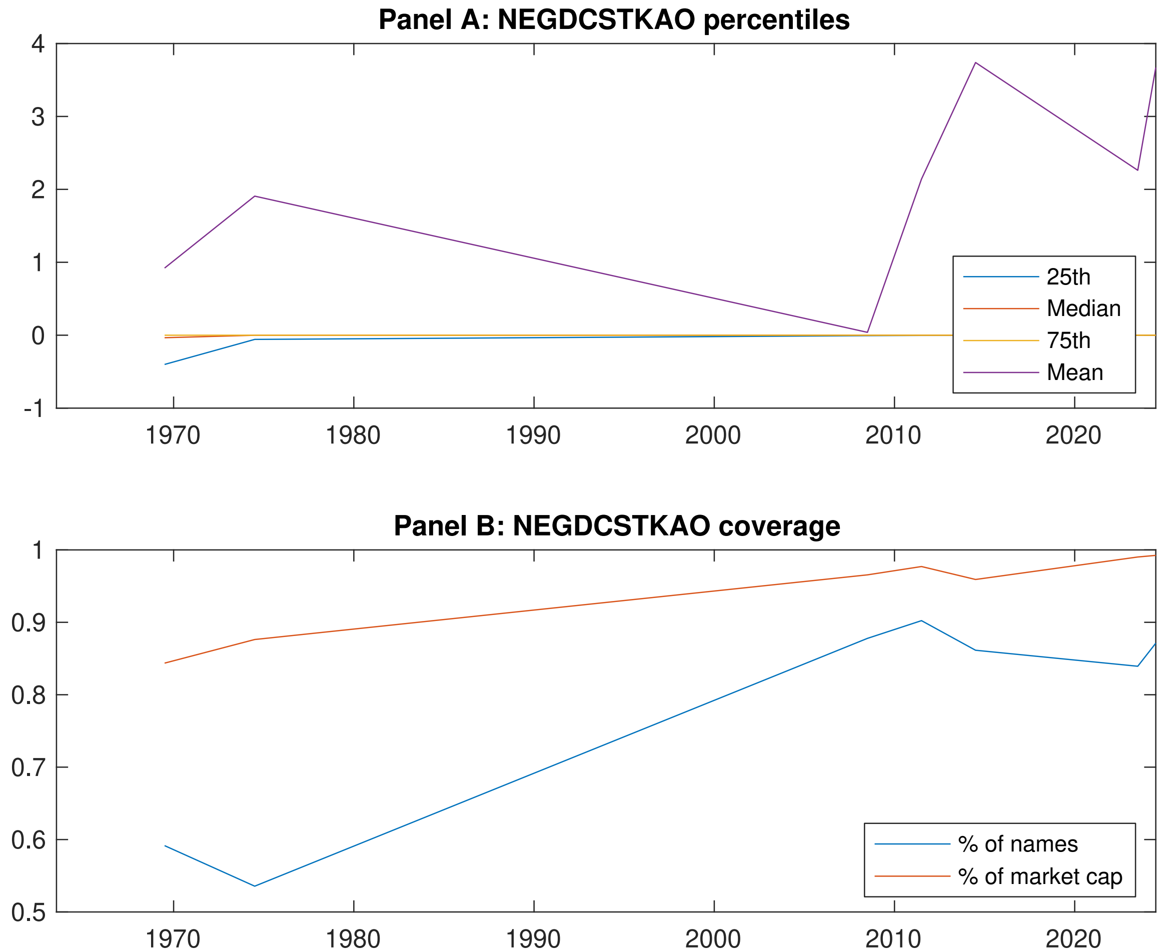


Figure 1: Times series of EFI percentiles and coverage.
This figure plots descriptive statistics for EFI. Panel A shows cross-sectional percentiles of EFI over the sample. Panel B plots the monthly coverage of EFI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46	0.50	0.61	0.64	0.76	0.30
	[2.77]	[2.77]	[3.42]	[3.96]	[4.79]	[4.18]
α_{CAPM}	-0.11	-0.12	-0.01	0.09	0.22	0.33
	[-2.22]	[-2.71]	[-0.14]	[1.86]	[4.86]	[4.48]
α_{FF3}	-0.10	-0.10	-0.01	0.04	0.18	0.28
	[-2.09]	[-2.35]	[-0.25]	[0.90]	[4.18]	[3.91]
α_{FF4}	-0.08	-0.08	0.01	0.04	0.16	0.25
	[-1.70]	[-1.85]	[0.16]	[0.83]	[3.71]	[3.36]
α_{FF5}	-0.11	-0.06	-0.00	-0.03	0.09	0.20
	[-2.29]	[-1.31]	[-0.09]	[-0.70]	[2.20]	[2.83]
α_{FF6}	-0.10	-0.04	0.01	-0.03	0.09	0.18
	[-1.97]	[-0.99]	[0.25]	[-0.59]	[2.01]	[2.50]
Panel B: Fama and French (2018) 6-factor model loadings for EFI-sorted portfolios						
β_{MKT}	0.96	1.02	1.05	0.99	0.98	0.02
	[80.35]	[93.35]	[96.41]	[91.53]	[96.19]	[1.08]
β_{SMB}	-0.01	0.02	-0.02	-0.07	-0.01	-0.01
	[-0.30]	[1.26]	[-1.29]	[-4.20]	[-0.90]	[-0.32]
β_{HML}	0.01	-0.02	-0.00	0.09	0.03	0.02
	[0.63]	[-0.85]	[-0.23]	[4.30]	[1.64]	[0.52]
β_{RMW}	0.08	-0.07	-0.03	0.12	0.14	0.06
	[3.48]	[-3.29]	[-1.50]	[5.53]	[7.21]	[1.81]
β_{CMA}	-0.08	-0.10	0.04	0.16	0.20	0.27
	[-2.29]	[-3.15]	[1.15]	[5.23]	[6.77]	[5.49]
β_{UMD}	-0.02	-0.02	-0.02	-0.01	0.01	0.03
	[-1.75]	[-1.79]	[-2.06]	[-0.61]	[0.95]	[1.74]
Panel C: Average number of firms (n) and market capitalization (me)						
n	778	622	527	635	693	
me (\$$10^6$)	1679	1426	2150	2224	2612	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EFI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30	0.33	0.28	0.25	0.20	0.18
			[4.18]	[4.48]	[3.91]	[3.36]	[2.83]	[2.50]
Quintile	NYSE	EW	0.48	0.55	0.46	0.39	0.32	0.26
			[7.13]	[8.49]	[7.83]	[6.61]	[5.69]	[4.78]
Quintile	Name	VW	0.33	0.34	0.30	0.27	0.25	0.22
			[4.38]	[4.53]	[4.00]	[3.49]	[3.27]	[2.94]
Quintile	Cap	VW	0.27	0.29	0.25	0.20	0.19	0.15
			[3.68]	[3.95]	[3.47]	[2.78]	[2.59]	[2.12]
Decile	NYSE	VW	0.29	0.30	0.24	0.20	0.22	0.19
			[3.34]	[3.48]	[2.80]	[2.34]	[2.48]	[2.16]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.26	0.28	0.24	0.22	0.18	0.17
			[3.66]	[3.94]	[3.37]	[3.08]	[2.56]	[2.39]
Quintile	NYSE	EW	0.29	0.37	0.29	0.25	0.14	0.12
			[3.92]	[5.20]	[4.37]	[3.82]	[2.33]	[1.99]
Quintile	Name	VW	0.29	0.31	0.26	0.24	0.22	0.21
			[3.87]	[4.10]	[3.57]	[3.30]	[3.05]	[2.87]
Quintile	Cap	VW	0.23	0.25	0.21	0.19	0.17	0.15
			[3.17]	[3.46]	[2.98]	[2.60]	[2.37]	[2.11]
Decile	NYSE	VW	0.24	0.25	0.20	0.18	0.18	0.16
			[2.85]	[2.91]	[2.30]	[2.07]	[2.06]	[1.91]

Table 3: Conditional sort on size and EFI

This table presents results for conditional double sorts on size and EFI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFI and short stocks with low EFI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EFI Quintiles					EFI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.49	0.72	0.84	0.95	1.01	0.54	0.61	0.53	0.49	0.35	0.33
		[1.88]	[2.85]	[3.35]	[3.79]	[4.49]	[5.56]	[6.36]	[5.92]	[5.34]	[4.09]	[3.75]
	(2)	0.57	0.73	0.82	0.96	0.97	0.40	0.46	0.34	0.30	0.27	0.24
		[2.48]	[3.07]	[3.57]	[4.39]	[4.48]	[4.45]	[5.18]	[4.22]	[3.61]	[3.29]	[2.87]
	(3)	0.60	0.60	0.76	0.87	0.90	0.30	0.33	0.25	0.21	0.18	0.16
		[2.97]	[2.82]	[3.54]	[4.30]	[4.61]	[3.75]	[4.13]	[3.25]	[2.71]	[2.35]	[1.99]
	(4)	0.49	0.62	0.81	0.76	0.82	0.33	0.38	0.29	0.28	0.12	0.13
		[2.54]	[3.14]	[4.11]	[4.05]	[4.56]	[4.17]	[4.76]	[3.97]	[3.75]	[1.75]	[1.82]
	(5)	0.46	0.52	0.45	0.53	0.73	0.27	0.28	0.25	0.21	0.23	0.20
		[2.82]	[2.98]	[2.54]	[3.25]	[4.63]	[3.13]	[3.17]	[2.83]	[2.33]	[2.59]	[2.23]

Panel B: Portfolio average number of firms and market capitalization

		EFI Quintiles					EFI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	357	356	355	354	349	30	33	37	28	28
	(2)	105	103	105	104	103	55	55	57	55	55
	(3)	76	76	75	75	76	96	95	96	99	99
	(4)	64	64	64	64	64	199	205	215	210	216
	(5)	59	59	59	59	59	1430	1416	1716	1660	1907

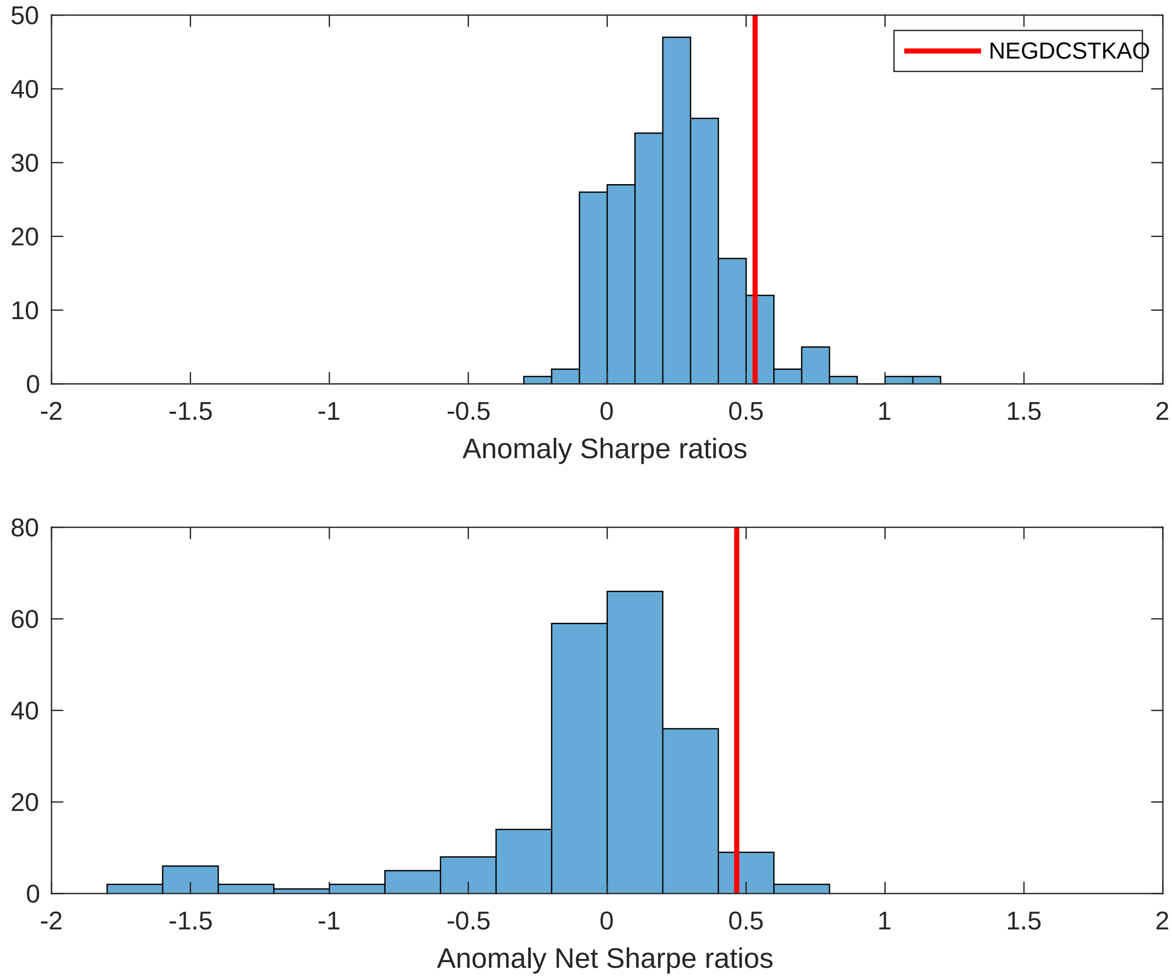


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

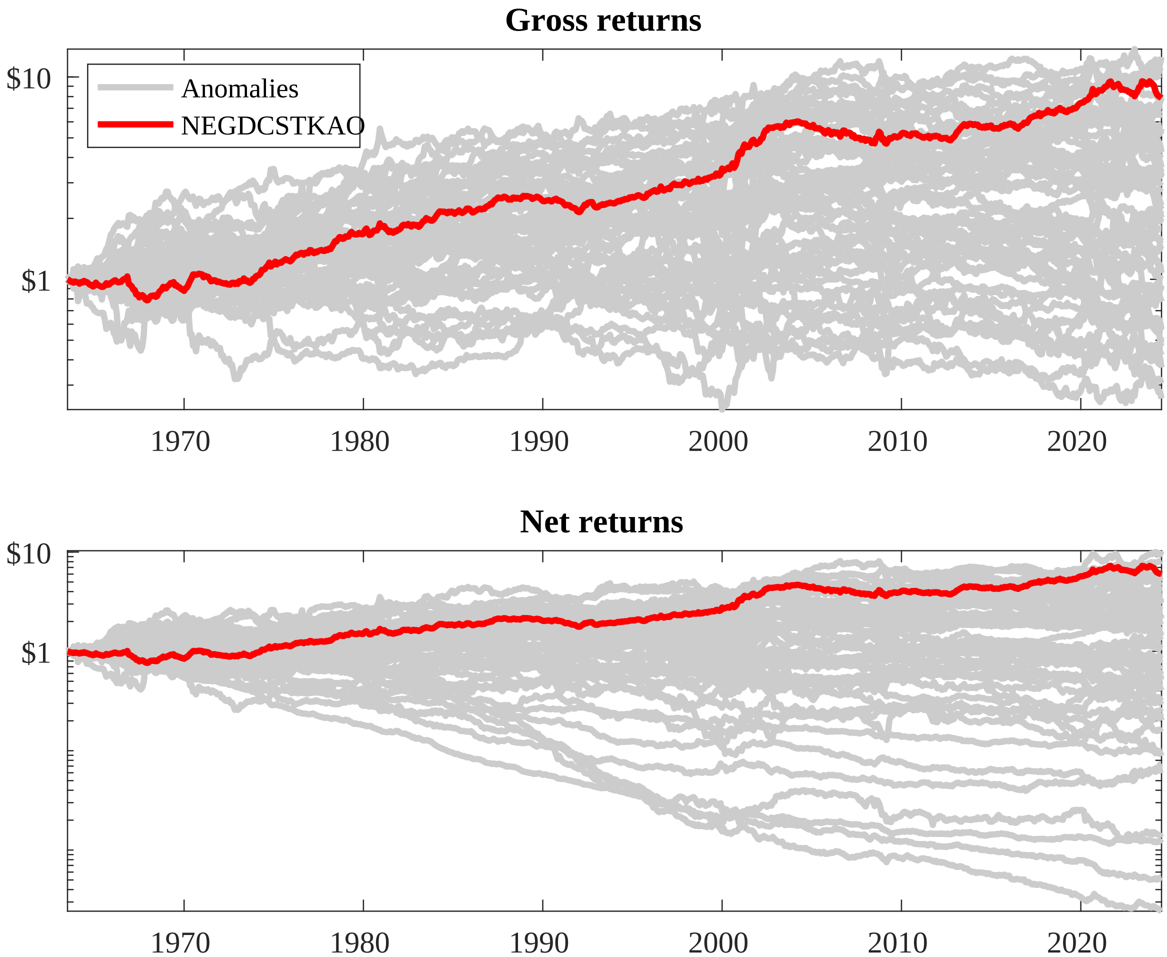
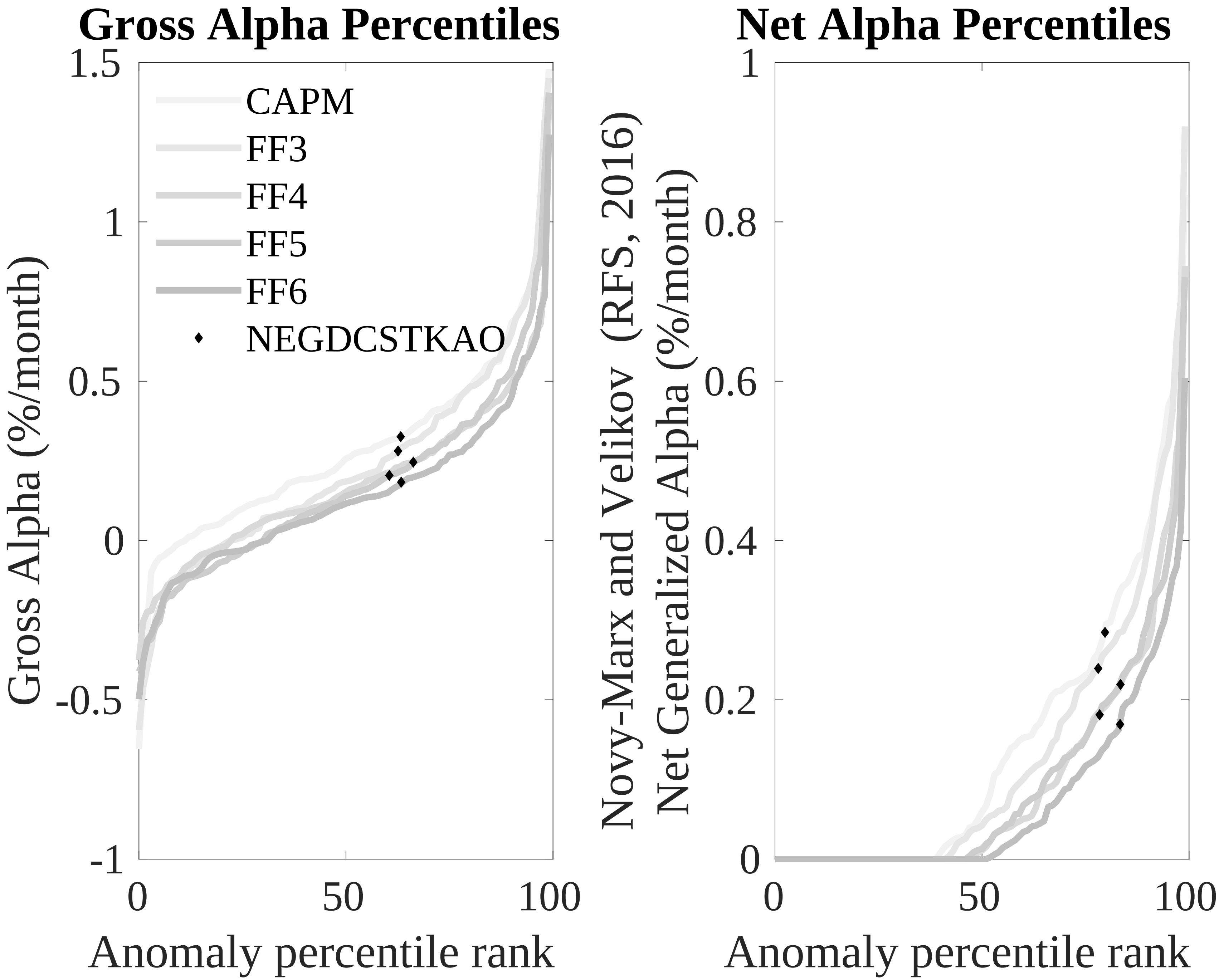
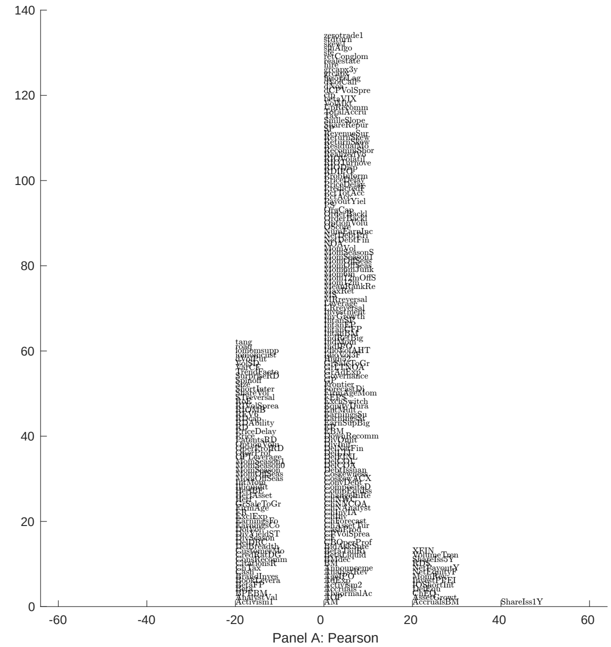


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
0
50
100
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKAO
Gross Alpha (%/month)
Anomaly percentile rank
0.8
0.6
0.4
0.2
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)



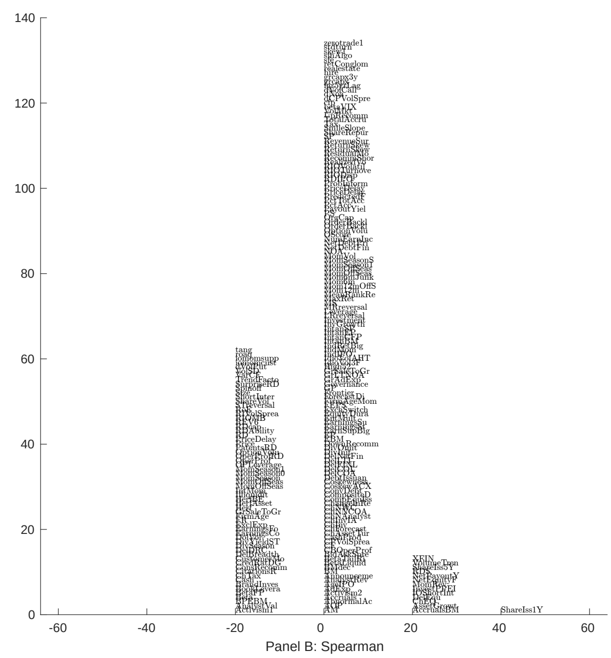


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

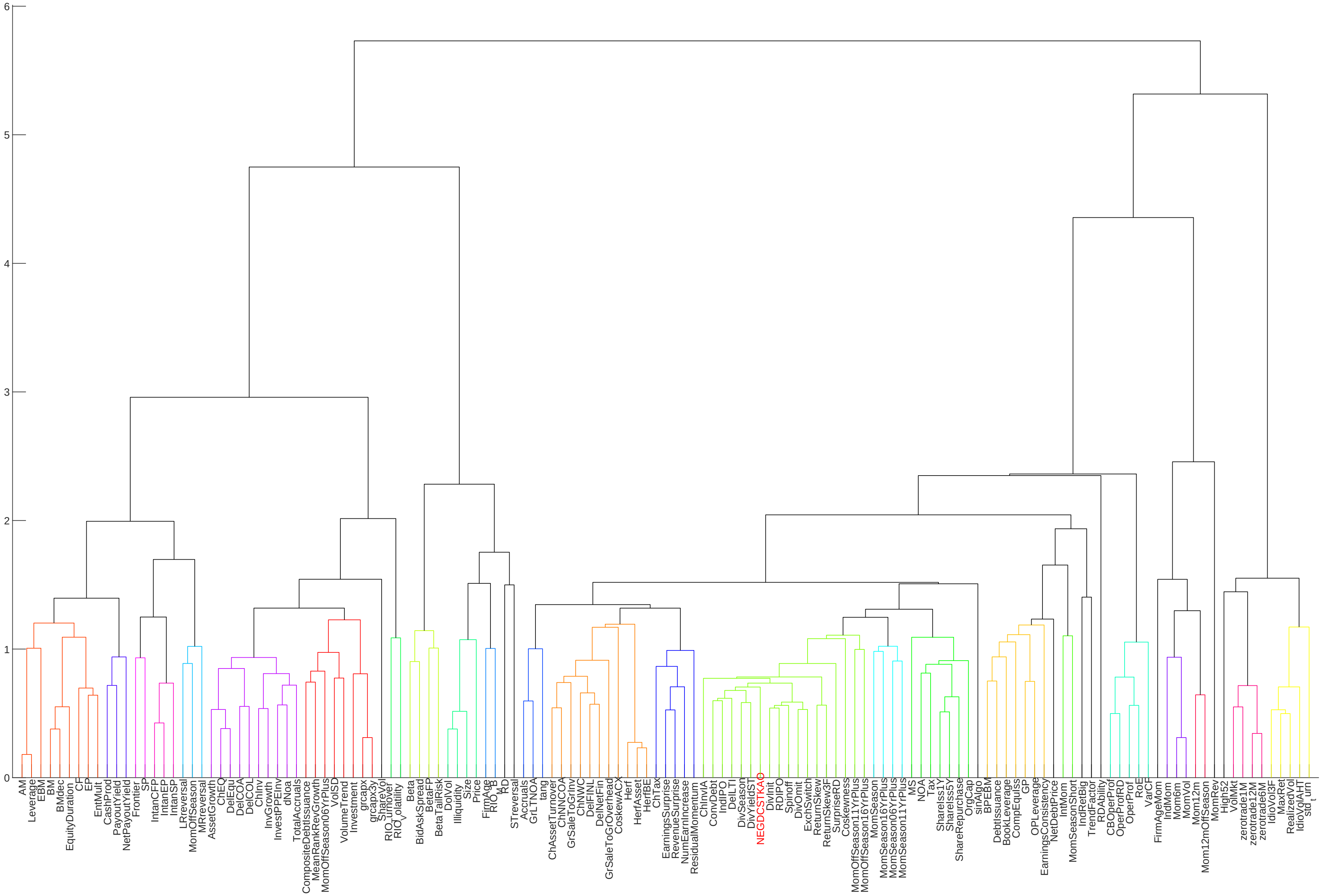


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

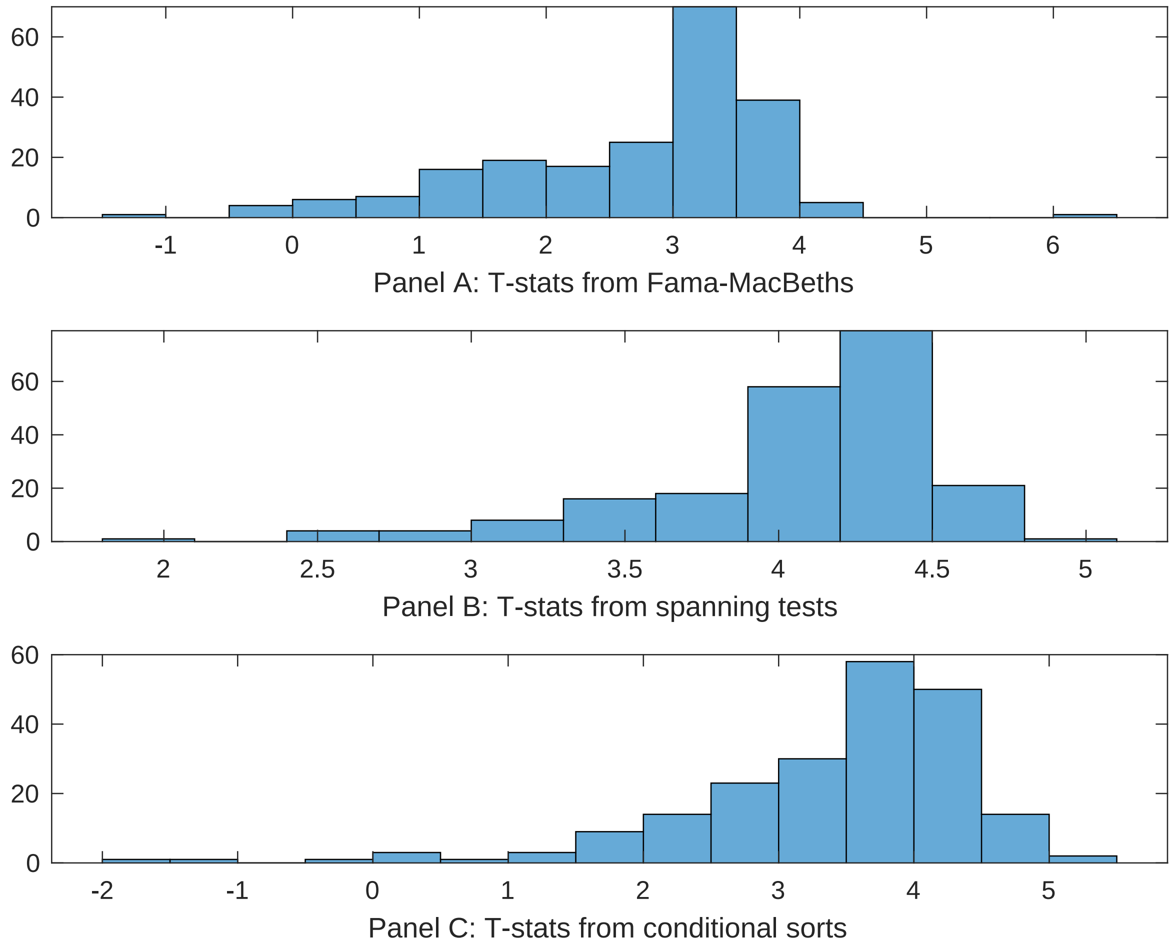


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI} EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFI} EFI_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.13	0.17	0.12	0.13	0.12
	[6.05]	[5.43]	[6.43]	[6.98]	[5.72]	[5.35]	[4.59]
EFI	0.62	0.27	0.55	0.51	0.74	0.59	0.27
	[2.43]	[0.91]	[2.23]	[1.81]	[2.55]	[2.05]	[0.85]
Anomaly 1	0.14						0.89
	[6.54]						[4.06]
Anomaly 2		0.19					0.13
		[1.66]					[2.74]
Anomaly 3			0.28				0.28
			[4.97]				[6.16]
Anomaly 4				0.41			-0.16
				[3.51]			[-1.11]
Anomaly 5					0.12		0.13
					[3.31]		[2.13]
Anomaly 6						0.16	-0.14
						[2.60]	[-1.77]
# months	715	715	715	720	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16	0.21	0.11	0.16	0.18	0.22	0.17
	[2.33]	[2.91]	[1.54]	[2.35]	[2.48]	[2.78]	[2.24]
Anomaly 1	27.62						10.02
	[7.78]						[2.06]
Anomaly 2		15.85					10.48
		[5.74]					[2.82]
Anomaly 3			28.29				19.07
			[7.54]				[3.96]
Anomaly 4				26.48			26.78
				[6.93]			[4.51]
Anomaly 5					12.11		-8.77
					[3.38]		[-1.61]
Anomaly 6						8.82	-13.22
						[2.50]	[-3.02]
mkt	2.77	4.55	3.36	2.95	1.86	3.50	3.65
	[1.66]	[2.60]	[2.00]	[1.75]	[1.08]	[1.83]	[1.97]
smb	1.33	2.80	2.13	-0.60	0.13	6.56	0.43
	[0.55]	[1.12]	[0.88]	[-0.25]	[0.05]	[2.15]	[0.15]
hml	2.21	-3.19	5.21	-0.29	1.36	2.92	0.41
	[0.69]	[-0.92]	[1.61]	[-0.09]	[0.41]	[0.84]	[0.12]
rmw	-8.79	-3.06	-3.97	7.41	7.27	6.05	-0.29
	[-2.32]	[-0.83]	[-1.12]	[2.26]	[2.15]	[1.46]	[-0.06]
cma	9.37	13.15	12.80	0.89	14.35	12.74	-13.88
	[1.80]	[2.47]	[2.53]	[0.15]	[2.36]	[2.24]	[-2.10]
umd	1.93	4.45	1.34	2.73	3.50	3.93	2.69
	[1.16]	[2.63]	[0.80]	[1.63]	[2.04]	[2.18]	[1.55]
# months	728	728	728	732	732	630	626
$\bar{R}^2(\%)$	16	13	16	15	10	8	18

Panel A: Average rank

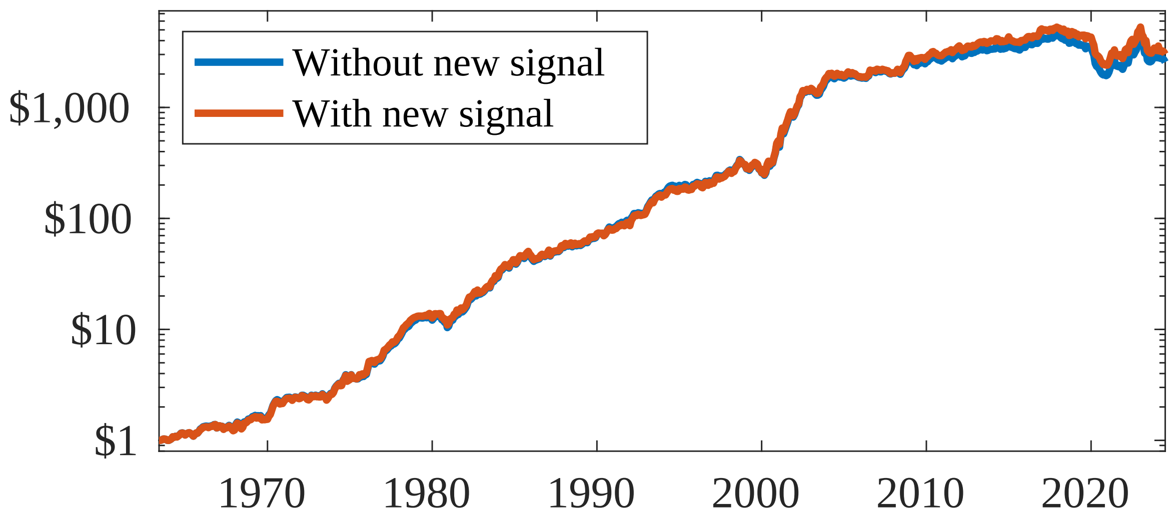


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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